

LC 1615 (Maximal Network Rank)

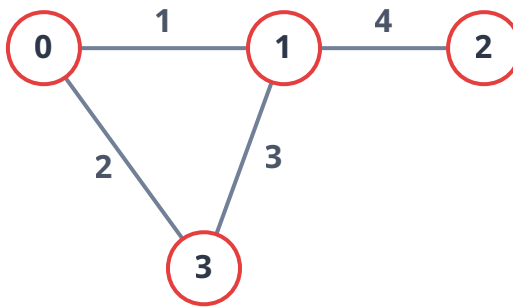
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Problem Description

- n cities, bidirectional roads
- network rank of 2 different cities is total number of directly connected roads to either city, if connected to both count only as 1
- need to return max network rank out of all pairs

Example



choose (0, 1) → total o/p = 4

if (1, 2) was selected → o/p = 3 {1, 4, 3}

Logic & Intuition

basically network rank of any 2 cities is

degree (city 1) + degree (city 2)

and if they are directly connected subtract 1, as road counted twice

C++ Solution

```
class Solution {
public:
    int maximalNetworkRank(int n, vector<vector<int>>& roads) {

        vector<vector<int>> adj(n);
        vector<int> deg(n,0);

        for(auto &it:roads){
            int u=it[0];
            int v=it[1];
            adj[u].push_back(v);
            adj[v].push_back(u);
            deg[u]++;
            deg[v]++;
        }

        int ans=INT_MIN;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){

                int currans=deg[i] + deg[j];

                if(count(adj[i].begin(),adj[i].end(),j)){
                    currans--;
                }
                //agar dono connected he to same road share krenge isliye -1

                ans=max(ans,currans);
            }
        }

        return ans;
    }
};
```